AWS Certified
Cloud Practitioner

(CLF-C02)

In-Depth Study Notes — All Four Exam Domains

A complete beginner-friendly reference covering Cloud Concepts, Security & Compliance, Cloud Technology & Services, and Billing, Pricing & Support — built to match the official exam guide weightings.

65 questions | 90 minutes | Pass mark: 700 / 1000

Exam Overview

The AWS Certified Cloud Practitioner (CLF-C02) is AWS's foundational certification. It tests a high-level understanding of AWS Cloud, its core services, security, architecture, pricing, and support — not hands-on engineering skill.

Domain	Weight	Focus
1. Cloud Concepts	24%	Value proposition, deployment models, Well-Architected Framework, migration
2. Security & Compliance	30%	Shared Responsibility Model, IAM, security services, compliance
3. Cloud Technology & Services	34%	Compute, storage, database, networking, AI/ML, deployment methods
4. Billing, Pricing & Support	12%	Pricing models, cost tools, support plans

Format: ~65 questions (multiple-choice and multiple-response), 90 minutes, $100 USD. 15 questions are unscored and not identifiable. Scoring is compensatory — you only need to pass overall, not each domain.

Exam tip: Technology (34%) + Security (30%) = 64% of the exam. If you're short on time, prioritize these two domains first.

DOMAIN 1: CLOUD CONCEPTS

Exam weight: 24%

1.1 What Is Cloud Computing?

Cloud computing is the on-demand delivery of IT resources (compute, storage, databases, networking, etc.) over the internet with pay-as-you-go pricing, instead of buying and maintaining physical data centers.

Deployment Models

- **Cloud (public):** Resources hosted entirely on AWS's infrastructure.
- **On-premises (private cloud):** Resources hosted in your own data center, sometimes virtualized like a cloud.
- **Hybrid:** A mix — some resources on AWS, some on-premises, connected together (e.g. via VPN or Direct Connect).

Six Advantages of Cloud Computing (AWS's official list — memorize this)

- Trade upfront expense for variable expense (CapEx → OpEx — pay only for what you use).
- Benefit from massive economies of scale (AWS's bulk purchasing lowers your per-unit cost).
- Stop guessing capacity (scale up or down as needed instead of over/under-provisioning).
- Increase speed and agility (new resources in minutes, not weeks).
- Stop spending money running and maintaining data centers (focus on your business, not hardware).
- Go global in minutes (deploy in multiple AWS Regions worldwide almost instantly).

Exam tip: Questions often describe one of these six benefits in a business scenario and ask you to identify which advantage it demonstrates.

1.2 AWS Global Infrastructure

- **Region:** A physical geographic location (e.g. us-east-1) containing multiple, isolated Availability Zones. Choose a region for compliance, latency, cost, and service availability.
- **Availability Zone (AZ):** One or more discrete data centers within a Region, each with independent power, cooling, and networking. Use multiple AZs for high availability.
- **Edge Location / Points of Presence:** Sites used by CloudFront (CDN) to cache content closer to end users, reducing latency.
- **Local Zones:** Extend a Region closer to large population/industry centers for very low-latency use cases.
- **Wavelength Zones:** AWS infrastructure embedded within telecom 5G networks for ultra-low-latency apps.

Exam tip: Design for at least 2 AZs for high availability. A single Region always has 2+ AZs.

1.3 AWS Well-Architected Framework

A framework of best practices organized into six pillars, used to evaluate and improve cloud architectures.

- **Operational Excellence:** Run and monitor systems, continually improve processes.
- **Security:** Protect data, systems, and assets.
- **Reliability:** Recover from failures and meet demand.

- **Performance Efficiency:** Use resources efficiently and adapt as needs change.
- **Cost Optimization:** Avoid unnecessary costs.
- **Sustainability:** Minimize environmental impact.

1.4 Cloud Adoption Framework (CAF) and Migration

The AWS CAF helps organizations plan and structure their cloud adoption journey across six perspectives: Business, People, Governance, Platform, Security, and Operations.

The 6 R's of Migration

- **Rehost** — "lift and shift"; move as-is with minimal changes.
- **Replatform** — "lift, tinker, and shift"; small optimizations without changing core architecture.
- **Repurchase** — move to a different product, often SaaS (e.g. switch to a cloud-based CRM).
- **Refactor / Re-architect** — redesign the application to be cloud-native.
- **Retire** — decommission resources no longer needed.
- **Retain** — keep as-is on-premises (do nothing, for now).

1.5 Key Cloud Concepts Vocabulary

- **Elasticity:** Automatically growing/shrinking resources to match demand.
- **Scalability:** Ability to increase resources to handle greater load (vertical = bigger instance; horizontal = more instances).
- **Agility:** Ability to rapidly develop, test, and launch applications.
- **High Availability:** Minimal downtime, typically via redundancy across AZs.
- **Fault Tolerance:** System keeps working even when a component fails.
- **Disaster Recovery:** Strategies (backup, pilot light, warm standby, multi-site) to recover from major failure.

1.6 Extra Topics Seen in Real Practice Exams (not just definitions)

The concepts below are commonly tested with scenario-style questions rather than simple "what is X" definitions. Each one below shows the KIND of question you'll actually face.

AWS Cloud Design Principles

Questions test whether you can pick the correct cloud-native design principle out of a mixed list.

Sample question: "Which of the following are AWS cloud design principles?" (Options mix real principles with distractors like "reserved capacity instead of on-demand" or "servers instead of managed services".)

Correct principles to know: Design for failure (treat resources as disposable, not fixed servers), loose coupling over tight coupling, implement elasticity, and use managed/serverless services instead of self-managing servers wherever possible. A distractor like "reserved capacity instead of on-demand" is WRONG because cloud favors dynamic provisioning based on real need, not fixed capacity.

Multi-AZ vs. Multi-Region (don't confuse the *benefit* each gives)

Sample question: "Which of the following is a benefit of running an application in multiple Availability Zones?" — with distractors like "reduces response time for global users" or "increases available compute capacity."

Correct answer logic: Multiple AZs primarily increase **availability/resilience** (if one AZ fails, your app keeps running from the others). Reducing latency for global users is what multiple *Regions* or CloudFront edge locations do — not what multi-AZ does. Don't mix these two benefits up.

AWS Migration Acceleration Program (MAP)

Sample question: "As part of the AWS Migration Acceleration Program (MAP), what does AWS provide to accelerate enterprise adoption of AWS?" (Options mix real MAP components with unrelated services like Amazon Athena or Amazon Pinpoint used as distractors.)

What to know: MAP gives access to **AWS Partners** (consulting/technology partners with migration expertise) and **AWS Professional Services** (AWS's own team that works directly with you on migration). It is a business/consulting program, not a technical service — so any specific AWS service listed as an answer option is almost always a distractor.

Migration Services You Must Tell Apart

Service	What it does
AWS Application Migration Service (MGN)	Automates "lift-and-shift" rehosting — continuously replicates on-prem servers to AWS and launches them as ready-to-go EC2 instances.
AWS Database Migration Service (DMS)	Migrates databases specifically (homogeneous or heterogeneous engines) to AWS with minimal downtime.
AWS Migration Hub	A single dashboard to track the progress of migrations happening across multiple AWS and partner migration tools.
AWS Migration Evaluator	Used BEFORE migrating — builds a data-driven cost/business case comparing on-premises costs vs. running the same workload on AWS.

Sample question: "What tool can a company use to perform a cost-benefit analysis of moving to the AWS Cloud?" → This is **Migration Evaluator**, not Cost Explorer or Pricing Calculator (those analyze AWS costs you're already incurring or estimate a specific AWS build, not a full on-prem-vs-cloud business case).

Migration Strategy: Repurchase vs. Replatform (a frequent trap)

Sample question: "A company replaces its on-premises email system with Amazon WorkMail, a cloud-based SaaS solution. Which migration strategy does this describe?" → This is **Repurchase** — swapping to a different (often SaaS) product entirely. It is NOT Replatform, which means making small optimizations while keeping the same basic application (e.g., moving a self-hosted database to Amazon RDS but keeping the same app).

DOMAIN 2: SECURITY AND COMPLIANCE

Exam weight: 30%

2.1 The Shared Responsibility Model (the single most tested topic)

Security is a shared duty between AWS and the customer.

AWS is responsible for "Security OF the Cloud"

- Physical security of data centers
- Global infrastructure (Regions, AZs, edge locations)
- Hardware, networking, and host operating system/virtualization layer

Customer is responsible for "Security IN the Cloud"

- Data encryption and protection
- Identity and Access Management (IAM) configuration
- Operating system patching, firewall/security group configuration (for services like EC2)
- Network traffic protection and client-side data

Exam tip: For managed/serverless services (e.g. S3, Lambda, RDS), AWS takes on more responsibility (patching the underlying OS) — but you're always responsible for your data and access permissions, no matter the service.

2.2 Identity and Access Management (IAM)

- **Root user:** Created with the account; has full access. Should be locked down with MFA and used only for account-level tasks — never for daily work.
- **IAM Users:** Individual identities with their own credentials.
- **IAM Groups:** Collections of users, used to apply permissions collectively.
- **IAM Roles:** Temporary permissions assumed by users, applications, or AWS services (e.g. an EC2 instance assuming a role to access S3). Preferred over hardcoding credentials.
- **IAM Policies:** JSON documents that define permissions (allow/deny actions on resources).
- **MFA (Multi-Factor Authentication):** Extra layer of login security — enable for root and all privileged users.
- **Principle of Least Privilege:** Grant only the permissions needed, nothing more.
- **IAM Identity Center (successor to AWS SSO):** Centralized access management across multiple AWS accounts.

2.3 Key Security, Detection, and Compliance Services

Service	What it does
AWS KMS	Create and manage encryption keys used to encrypt data across AWS services.
AWS Shield	Managed DDoS protection. Standard is free/automatic; Advanced offers enhanced protection and cost protection.

Service	What it does
AWS WAF	Web Application Firewall — filters malicious web traffic (e.g. SQL injection, XSS) in front of apps.
Amazon GuardDuty	Intelligent threat detection that continuously monitors for malicious activity using ML.
Amazon Inspector	Automated security assessments for vulnerabilities in EC2, containers, and Lambda.
Amazon Macie	Uses ML to discover and protect sensitive data (like PII) stored in S3.
AWS Artifact	Self-service portal for AWS compliance reports and agreements (e.g. ISO, PCI-DSS, SOC).
AWS CloudTrail	Logs and tracks API calls / user activity across your AWS account ("who did what, when").
AWS Config	Tracks resource configurations over time and checks compliance against rules.
Security Groups	Virtual firewall at the instance level; stateful; only supports "allow" rules.
Network ACLs (NACLs)	Virtual firewall at the subnet level; stateless; supports both "allow" and "deny" rules.
AWS Organizations	Centrally manage and govern multiple AWS accounts; apply Service Control Policies (SCPs).
AWS Secrets Manager	Securely store, rotate, and retrieve credentials like database passwords and API keys.
Amazon Cognito	Manages user sign-up, sign-in, and access control for web/mobile apps.

Exam tip: Know the difference: Security Groups = instance-level, stateful, allow-only. NACLs = subnet-level, stateless, allow AND deny. This distinction is commonly tested.

2.4 Compliance

- AWS complies with many global standards: ISO 27001, PCI-DSS, HIPAA, GDPR, SOC 1/2/3, FedRAMP, etc.
- You are still responsible for ensuring *your own* application/data usage is compliant — AWS provides the compliant infrastructure, not automatic compliance for your workload.
- AWS Artifact is where you download these compliance reports and agreements (like the BAA for HIPAA).

2.5 Extra Topics Seen in Real Practice Exams (not just definitions)

Root User Protection — know the FULL best-practice list

Sample question: "Which of the following strategies helps protect your AWS root account?" — with a tricky distractor like "apply MFA for the root account AND use it for all of your work."

Why the distractor is wrong: Enabling MFA on root is correct, but the second half — *using the root account for daily work* — breaks best practice. The full checklist is: enable MFA on root, **delete root access keys entirely** if you don't strictly need them, never use root for everyday tasks (create an IAM user/role instead), and share credentials with no one.

Amazon Connect (a real service, but a common wrong answer)

Sample question: "Which of the following can be used to enable Virtual Multi-Factor Authentication?" — Amazon Connect is listed as a distractor option alongside AWS CLI and IAM.

What to know: Amazon Connect is a cloud-based **contact center / customer service call center** service — it has nothing to do with authentication or MFA. It shows up often as a distractor in Security domain questions precisely because its name sounds security-related. Don't fall for it.

Directory Service vs. IAM Identity Center (frequently swapped)

Sample question: "Which AWS service allows a company to connect their AWS resources with an on-premises Microsoft Active Directory, enabling users to sign in with existing corporate credentials?"

Correct answer: AWS Directory Service (Managed Microsoft AD) — connects/extends an existing on-prem Active Directory into AWS. **IAM Identity Center** (formerly AWS SSO) is different: it provides single sign-on access *across multiple AWS accounts and business apps*, and can use Active Directory as an identity source, but it is not itself the AD-connection service being described here.

Certificate Manager vs. IAM for SSL Certificates

Sample question: "A developer needs an SSL certificate for an e-commerce website to use HTTPS. Which services can deploy the required SSL server certificates?" (multi-select)

Correct answers: AWS Certificate Manager (ACM) is the primary way to provision/manage/deploy SSL/TLS certificates for services like CloudFront and ELB. **IAM** can also store and deploy SSL certificates for services ACM doesn't support yet. Route 53 (DNS) and AWS Directory Service are distractors with no certificate role.

Config vs. CloudTrail vs. Inspector — the classic 3-way mix-up

Sample question: "Which AWS service allows users to identify the changes made to a resource over time?" → **AWS Config** (tracks configuration changes/history). Compare to: "Which service enables risk auditing by continuously logging account activity, including console and SDK actions?" → **CloudTrail** (logs WHO did WHAT). And: "Which service performs automated vulnerability/network assessments of EC2 instances?" → **Amazon Inspector** (scans for vulnerabilities). All three commonly appear together as options in the same question — know the one-word difference: Config = configuration history, CloudTrail = activity log, Inspector = vulnerability scan.

Change Management / Reliability — AWS Config + CloudTrail together

Sample question: "What are the change management tools that help AWS customers audit and monitor all resource changes in their environment?" (Select TWO) → **AWS CloudTrail** and **AWS Config** together are the standard "correct pair" answer for change-tracking/auditing questions.

DOMAIN 3: CLOUD TECHNOLOGY AND SERVICES

Exam weight: 34%

This is the largest domain. You need only a *basic* grasp of what each service does and when to use it — not deep configuration knowledge.

3.1 Ways to Interact with AWS

- **AWS Management Console:** Web-based graphical interface.
- **AWS CLI:** Command-line interface for scripting.
- **AWS SDKs:** Libraries to call AWS APIs from code (Python, Java, JS, etc.).
- **Infrastructure as Code:** AWS CloudFormation (JSON/YAML templates) or AWS CDK.

3.2 Compute Services

Service	What it does
Amazon EC2	Resizable virtual servers (VMs). You choose instance type, OS, storage. Pay per second/hour.
AWS Lambda	Serverless, run code without managing servers; pay only for compute time used; auto-scales.
Amazon ECS	Container orchestration service to run Docker containers.
Amazon EKS	Managed Kubernetes service for container orchestration.
AWS Fargate	Serverless compute engine for containers (used with ECS/EKS) — no server management.
AWS Elastic Beanstalk	PaaS — upload your code, AWS handles provisioning, load balancing, scaling automatically.
Amazon Lightsail	Simplified VPS for simple websites/apps with predictable low pricing.
AWS Outposts	Extends AWS infrastructure and services to your on-premises data center.
AWS Batch	Runs large-scale batch computing jobs efficiently.

Exam tip: EC2 = you manage the server. Lambda = fully serverless, no server management at all. Elastic Beanstalk = AWS manages the infrastructure for you, but it's still running on EC2 under the hood.

3.3 Storage Services

Service	What it does
Amazon S3	Object storage for any amount of data; 11 nines durability; storage classes: Standard, Intelligent-Tiering, Standard-IA, One Zone-IA, Glacier (archival, slow/cheap retrieval).

Service	What it does
Amazon EBS	Block storage volumes attached to a single EC2 instance (like a virtual hard drive).
Amazon EFS	Managed, scalable file storage (NFS) that can be shared across many EC2 instances at once.
Amazon FSx	Managed file systems for Windows File Server or Lustre (high-performance computing).
AWS Storage Gateway	Hybrid storage connecting on-premises environments with AWS cloud storage.
AWS Snow Family	Physical devices (Snowcone, Snowball, Snowmobile) to migrate very large amounts of data offline.
AWS Backup	Centralized, automated backup service across multiple AWS services.

Exam tip: S3 = object storage (files/objects, accessed via API/URL). EBS = block storage (attached like a disk to ONE EC2 instance). EFS = file storage (shared by MANY instances at once). This trio is a classic exam question.

3.4 Database Services

Service	What it does
Amazon RDS	Managed relational database (MySQL, PostgreSQL, SQL Server, Oracle, MariaDB). AWS handles patching, backups.
Amazon Aurora	AWS's own high-performance, MySQL/PostgreSQL-compatible relational database; faster and more scalable than standard RDS engines.
Amazon DynamoDB	Fully managed, serverless NoSQL key-value/document database; single-digit millisecond performance at any scale.
Amazon Redshift	Fully managed data warehouse for large-scale analytics and business intelligence.
Amazon ElastiCache	Managed in-memory caching (Redis or Memcached) to speed up applications.
Amazon Neptune	Managed graph database for highly connected data (social networks, recommendation engines).
Amazon DocumentDB	Managed document database compatible with MongoDB.
AWS Database Migration Service (DMS)	Migrates databases to AWS with minimal downtime.

3.5 Networking Services

Service	What it does
Amazon VPC	Your own logically isolated virtual network within AWS, with subnets, route tables, and gateways.
Subnets	Segments of a VPC's IP range; public subnets route to the internet, private subnets do not.
Internet Gateway	Allows communication between a VPC and the public internet.

Service	What it does
NAT Gateway	Allows resources in a private subnet to reach the internet (outbound only) without being publicly reachable.
Amazon Route 53	Scalable DNS and domain registration service.
Amazon CloudFront	Content Delivery Network (CDN) — caches content at edge locations for low-latency delivery.
Elastic Load Balancing (ELB)	Distributes incoming traffic across multiple targets (EC2 instances, containers, etc.).
AWS Direct Connect	Dedicated private network connection from on-premises to AWS (bypasses public internet).
AWS VPN	Encrypted connection between on-premises and AWS over the public internet.
AWS Auto Scaling	Automatically adds/removes EC2 instances based on demand or schedules.

3.6 Management, Monitoring, and Deployment Tools

Service	What it does
Amazon CloudWatch	Monitoring service — collects metrics, logs, sets alarms on AWS resources and applications.
AWS CloudTrail	Audit log of API calls/account activity (who did what).
AWS CloudFormation	Infrastructure as Code — define and provision AWS resources using templates.
AWS Systems Manager	Operational hub for managing EC2/on-prem servers (patching, automation, parameter store).
AWS Trusted Advisor	Automated recommendations across cost optimization, performance, security, fault tolerance, and service limits.
AWS Organizations	Manage multiple AWS accounts centrally; consolidated billing; Service Control Policies.

Exam tip: CloudWatch = performance/health monitoring (metrics, alarms). CloudTrail = activity/audit logging (who did what). Don't mix these up.

3.7 Artificial Intelligence and Machine Learning

Service	What it does
Amazon SageMaker	Fully managed platform to build, train, and deploy custom ML models.
Amazon Rekognition	Image and video analysis (object/face detection).
Amazon Comprehend	Natural language processing — extracts insights and sentiment from text.
Amazon Polly	Converts text to lifelike speech.
Amazon Transcribe	Converts speech to text.

Service	What it does
Amazon Translate	Language translation.
Amazon Lex	Build conversational chatbots (same tech as Alexa).
Amazon Bedrock	Access foundation/generative AI models from AWS and third parties via API.
Amazon Q	Generative AI-powered assistant for work tasks and for developers using AWS.

Exam tip: For AI/ML services, you just need to match the service name to its one-line purpose — no technical depth required.

3.8 Application Integration

Service	What it does
Amazon SNS	Pub/sub messaging — sends notifications to multiple subscribers (email, SMS, apps).
Amazon SQS	Fully managed message queue — decouples application components.
Amazon EventBridge	Serverless event bus connecting applications with data from various sources.
Amazon API Gateway	Create, publish, and manage APIs at scale.

3.9 Analytics Services (appear often — just match name to purpose)

Service	What it does
Amazon Athena	Serverless, run SQL queries directly on data stored in S3 — no infrastructure to manage.
Amazon QuickSight	Business intelligence / dashboard and data-visualization service.
AWS Glue	Serverless ETL (extract, transform, load) service to prepare and catalog data for analytics.
Amazon EMR	Managed Hadoop/Spark big-data processing framework for large-scale data processing.
Amazon Kinesis	Real-time streaming data ingestion, processing, and analysis at scale.

Sample question: "A user needs a single location to see the status of S3 bucket permissions, whether root MFA is enabled, and whether security groups allow unrestricted access." QuickSight is a tempting wrong answer because it sounds like a "dashboard," but the actual right answer is **AWS Trusted Advisor** — QuickSight visualizes your OWN business/analytics data, it does not generate built-in AWS security checks.

3.10 Business Applications (often used as distractors in Security questions)

Service	What it does
Amazon Connect	Cloud-based contact center / customer service call center service.
Amazon Chime	Video conferencing and online meetings.
Amazon WorkMail	Managed business email and calendar (a SaaS replacement for on-prem Exchange).

Service	What it does
Amazon WorkDocs	Managed document storage and collaboration/sharing.
Amazon WorkSpaces	Managed virtual desktops (Desktop-as-a-Service).
Amazon AppStream 2.0	Streams desktop applications to a browser without installing them locally.
Amazon Pinpoint	Marketing communications — targeted email, SMS, and push notifications.
Amazon SES	Sends transactional/marketing emails at scale (Simple Email Service).

Exam tip: These names show up constantly as WRONG answers in Security and Technology questions because they sound technical. If a question is about MFA, IAM, encryption, or auditing, and one option is Connect/Chime/WorkMail/Pinpoint — it's almost certainly a distractor.

3.11 Extra Topics Seen in Real Practice Exams (not just definitions)

VPC Peering vs. Transit Gateway (a very common scaling question)

Sample question: "A company has hundreds of VPCs across multiple Regions worldwide. What service simplifies connection management among the VPCs?"

Correct answer: AWS Transit Gateway — a central hub that connects thousands of VPCs through ONE managed gateway. **VPC Peering** only creates individual one-to-one connections, which becomes unmanageable at scale (this is the classic wrong answer for "hundreds of VPCs" questions — peering doesn't scale, Transit Gateway does).

CloudFront Edge Locations vs. Global Accelerator (easy to confuse)

Sample question: "What does Amazon CloudFront use to distribute content to global users with low latency?" → **Edge Locations** (a global network of caching points of presence), NOT AWS Global Accelerator. Global Accelerator is a separate service that improves performance by routing traffic to the optimal *regional endpoint* using AWS's global network — it doesn't cache content like CloudFront does.

S3 Storage Classes — match the SCENARIO, not just the name

Questions almost always give you a scenario (access pattern + cost sensitivity + resiliency need) and ask which storage class fits — memorizing names alone won't help.

- **"Popular e-commerce site, stable/frequent access"** → S3 Standard (NOT Standard-IA — that's for infrequent access).
- **"Duplicate backups that could be recreated, rarely needed but must be fast when needed, lower resiliency OK"** → S3 One Zone-IA (single AZ = cheaper, lower resiliency).
- **"Compliance recordings only needed in a legal dispute, rarely if ever accessed"** → S3 Glacier Deep Archive (the cheapest class; retrieval takes hours).
- **"Unpredictable or changing access patterns"** → S3 Intelligent-Tiering (auto-moves data between tiers for you).

Exam tip: If a question emphasizes UNKNOWN/CHANGING access patterns → Intelligent-Tiering. If it emphasizes NEVER/RARELY accessed and cost is the top priority → Glacier Deep Archive. If it's a popular, frequently-hit website → Standard.

S3 Data Protection Features

Sample question: "Which of the following can be used to protect data at rest on Amazon S3?" (Select TWO) → **Versioning** (keeps multiple object versions, protects against accidental deletion) and **Permissions** (IAM/bucket policies controlling access) are correct. Deduplication, decryption, and "conversion" are distractors with no real protective function here.

RDS Multi-AZ vs. Read Replicas vs. Cross-Region Read Replicas

Sample question: "What feature of Amazon RDS helps create globally redundant databases?" → **Cross-Region Read Replicas** (asynchronously replicated copies in another Region, promotable to primary during a regional failure). Don't confuse with plain **Multi-AZ**, which protects against an *Availability Zone* failure within the SAME Region, not a full Region-wide outage.

Amazon Machine Image (AMI) vs. AppStream 2.0

Sample question: "Which AWS feature enables a user to launch a pre-configured EC2 instance?" → **Amazon Machine Image (AMI)** — a template with OS, app server, and code baked in. AppStream 2.0 is a distractor here since it streams desktop apps to a browser, it doesn't launch EC2 instances for you.

Instance Store — know its one key limitation

Sample question: "Which statement is true regarding the security of Amazon EC2?" — with a wrong option like "you should use instance store volumes to store login data."

What to know: Instance Store provides temporary, high-speed local storage that is **deleted the moment the instance stops or terminates** — never store anything you need to keep (like credentials/login data) on it. The correct answer to this kind of question is usually "regularly patch the OS and applications on your EC2 instances" (a customer responsibility under the Shared Responsibility Model).

Snowball Edge vs. S3 Transfer Acceleration vs. Direct Connect

Sample question: "A company needs to migrate a 100 TB on-premises database to AWS but has limited network bandwidth. Which service is MOST appropriate?" → **AWS Snowball Edge** (a physical device, holds up to ~80 TB, ships to AWS — ideal when bandwidth, not distance, is the bottleneck). **S3 Transfer Acceleration** instead speeds up uploads over the internet using CloudFront edge locations — it still depends on your network, so it's the wrong choice when bandwidth is the limiting factor.

DOMAIN 4: BILLING, PRICING, AND SUPPORT

Exam weight: 12%

4.1 AWS Pricing Fundamentals

Three fundamental pricing drivers: **Compute** (pay for what you use), **Storage** (pay for what you use), and **Data transfer OUT** (data transfer IN is generally free; transfer between services in the same Region is often free too).

EC2 Pricing Models

- **On-Demand:** Pay by the second/hour, no commitment; most flexible, most expensive per hour.
- **Reserved Instances:** 1 or 3-year commitment for a discount (up to ~72%); best for steady, predictable workloads.
- **Savings Plans:** Commit to a consistent $/hour spend for 1 or 3 years in exchange for lower prices — more flexible than Reserved Instances.
- **Spot Instances:** Bid on spare EC2 capacity for up to ~90% discount; AWS can reclaim the instance with short notice — best for fault-tolerant, flexible workloads (batch jobs, big data).
- **Dedicated Hosts:** Physical server dedicated to you — for compliance or licensing needs (e.g. bring-your-own-license software).

Exam tip: A scenario about "flexible, interruptible workloads at the lowest price" = Spot Instances. "Steady-state workload running 24/7 for a year" = Reserved Instance or Savings Plan.

4.2 Free Tier

- **12 Months Free:** Limited usage of certain services (e.g. some EC2/S3/RDS usage), free for the first 12 months after account creation.
- **Always Free:** Certain services/limits are free indefinitely (e.g. limited Lambda invocations).
- **Short-Term Trials:** Free trial for specific services for a limited time.

4.3 Cost Management Tools

Service	What it does
AWS Pricing Calculator	Estimate the cost of AWS services before you deploy them.
AWS Cost Explorer	Visualize, understand, and analyze your AWS costs and usage over time.
AWS Budgets	Set custom cost/usage budgets and receive alerts when thresholds are exceeded.
Cost and Usage Report (CUR)	The most detailed, granular breakdown of your AWS costs and usage.
Cost Allocation Tags	Tag resources to track/categorize costs by project, team, or department.
AWS Organizations (Consolidated Billing)	Combine usage across multiple linked accounts into a single bill, unlocking volume discounts.

4.4 AWS Support Plans

Plan	Best For	Key Features
Basic	All customers (free)	Customer service, documentation, forums, Trusted Advisor (core checks), Personal Health Dashboard
Developer	Testing/early-stage	Business-hours email support, general guidance, response in <24h
Business	Production workloads	24/7 phone/chat/email, full Trusted Advisor checks, <1h response for urgent issues
Enterprise On-Ramp	Growing orgs	Pooled TAM support, <30 min response for critical issues
Enterprise	Mission-critical	Dedicated Technical Account Manager (TAM), <15 min response for critical issues, proactive guidance

Exam tip: Only Business tier and above include full Trusted Advisor checks and 24/7 support. Only Enterprise (and On-Ramp) includes a Technical Account Manager (TAM).

4.5 Billing Alerts and Governance

- Use **AWS Budgets** to get notified via email/SNS before you overspend.
- Use **AWS Organizations** with **Service Control Policies (SCPs)** to restrict what member accounts can do.
- The **AWS Free Tier alert** notifies you as you approach free tier limits.

4.6 Extra Topics Seen in Real Practice Exams (not just definitions)

AWS Cost Anomaly Detection (distinct from Budgets)

Sample question: "A company wants to automatically detect unusual spending patterns and receive alerts when anomalies are found, without manually setting thresholds. Which service uses machine learning to do this?" → **AWS Cost Anomaly Detection** — NOT AWS Budgets. Budgets requires YOU to set a fixed spending threshold manually; Cost Anomaly Detection uses ML to learn your normal spending pattern and flag deviations automatically, with no threshold-setting required.

Migration Evaluator vs. Pricing Calculator vs. Cost Explorer (all sound similar — know the timing)

- **AWS Migration Evaluator** — used BEFORE migrating; builds a business case comparing on-prem vs AWS costs.
- **AWS Pricing Calculator** — used BEFORE building; estimates the cost of a planned AWS architecture.
- **AWS Cost Explorer** — used AFTER deployment; visualizes and analyzes costs you're already incurring.

Sample question: "What tool can a company use to perform a cost-benefit analysis of moving their entire on-premises data center to AWS?" → **Migration Evaluator** (this is the "before you even decide to migrate" tool — Cost Explorer and Budgets only look at AWS usage you already have).

What Actually Factors Into EC2 Cost (a "pick what does NOT belong" style question)

Sample question: "Which of the following is NOT a factor when estimating the cost of Amazon EC2?" (Select TWO) → Correct non-factors: **number of Security Groups** and **number of Hosted Zones** (a Route 53 concept) — neither has any cost impact on EC2. Real EC2 cost factors ARE: instance type, number of running instances, hours running, attached storage, and data transfer out, plus any allocated but *unused* Elastic IP addresses (AWS charges for Elastic IPs that are allocated but not actively attached to a running instance — a detail that surprises many beginners).

AWS Managed Services vs. Services YOU Manage Yourself

Sample question: "Which of the following are examples of AWS-managed services, where AWS handles the operational/maintenance burden?" (Select TWO, from a list mixing DynamoDB, EMR, VPC, IAM, and EC2)

Correct answer logic: DynamoDB and **EMR** are fully managed — AWS handles provisioning, patching, and scaling. **VPC, IAM, and EC2** are customer-configured constructs/resources — YOU define the subnets, policies, and OS patches. This same logic ("is AWS managing the internals, or am I?") applies across many billing/support questions about where operational overhead sits.

Support Plan Response Times — the numbers actually get tested

Sample question style: A scenario describes an urgent production issue and asks which support plan guarantees a certain response time.

- Basic / Developer: no guaranteed response time for "general guidance" beyond business-hours email (Developer).
- Business / Enterprise On-Ramp: as fast as <1 hour for "business-critical" issues.
- Enterprise: as fast as <15 minutes for "business-critical" issues, plus a named Technical Account Manager (TAM).

Exam tip: If a question mentions a dedicated Technical Account Manager (TAM), the answer is Enterprise (or Enterprise On-Ramp for a pooled TAM). If it just mentions 24/7 phone support with no TAM, that's Business.

Final Exam-Day Strategy

- Read the LAST sentence of a scenario question first — it usually reveals what's actually being asked.
- Watch for qualifier words: *MOST cost-effective*, *LEAST operational overhead*, *MOST secure*, *HIGHLY available* — these point directly to the right service.
- Eliminate the two obviously wrong answers first, then decide between the remaining two.
- For "Select TWO/THREE" questions — select exactly the number asked, no more, no less.
- Flag long/confusing questions and revisit at the end; you have ~1.4 minutes per question.
- There's no penalty for guessing — never leave a question blank.

Quick-Revision: Commonly Confused Pairs

Service	What it does
S3 vs EBS vs EFS	S3 = object storage (files via API). EBS = block storage for ONE EC2 instance. EFS = shared file storage for MANY instances.
Security Group vs NACL	Security Group = instance-level, stateful, allow-only. NACL = subnet-level, stateless, allow + deny.
CloudWatch vs CloudTrail vs Config	CloudWatch = performance monitoring & alarms. CloudTrail = who-did-what audit log. Config = resource configuration history over time.
RDS vs DynamoDB	RDS = managed relational (SQL) database. DynamoDB = serverless NoSQL key-value database.
Reserved vs Spot Instances	Reserved = steady, predictable workloads, big discount for commitment. Spot = flexible/interruptible workloads, deepest discount.
IAM User vs Role	User = long-term credentials for a person. Role = temporary credentials assumed by a user, app, or AWS service.
VPC Peering vs Transit Gateway	Peering = one-to-one VPC links, doesn't scale. Transit Gateway = central hub connecting hundreds/thousands of VPCs.
CloudFront vs Global Accelerator	CloudFront = caches content at edge locations (CDN). Global Accelerator = routes traffic to the best regional endpoint (no caching).
Directory Service vs IAM Identity Center	Directory Service = connects/hosts Active Directory. IAM Identity Center = SSO across multiple AWS accounts/apps.
Multi-AZ vs Cross-Region Read Replica	Multi-AZ = survives an AZ failure within one Region. Cross-Region Read Replica = survives a whole-Region outage.
Migration Evaluator vs Pricing Calculator vs Cost Explorer	Evaluator = pre-migration on-prem-vs-AWS business case. Calculator = estimate a planned build. Cost Explorer = analyze costs already incurred.
Amazon Connect (the trap)	It's a cloud contact-center / call-center service — NOT related to security, MFA, or networking, despite the name.

Good luck on your exam! Review this document 2-3 times, pair it with free official AWS practice questions and a full-length timed practice test, and focus extra time on Security (30%) and Technology (34%) since together they

make up nearly two-thirds of the exam.

MISTAKE REVIEW: PRACTICE EXAM ATTEMPT 1

This section covers every question in your results report that you got wrong or only partly right (29 questions). For each one you get the question, what you picked, the correct answer, why it is correct, and a one-line memory hook. After the cards there are comparison tables for the topics you slipped on, so you can revise the whole topic and not just that one question.

A. Patterns in your mistakes

Reading all 29 together, the errors are not random. They fall into four clear patterns:

- **Lookalike services swapped (most common).** Step Functions vs EventBridge, CodePipeline vs CodeDeploy, EC2 Instance Connect vs Session Manager, Cognito vs IAM Identity Center, Service Health vs Account Health dashboard, CloudWatch vs Trusted Advisor, Hardware MFA vs U2F key, Standard-IA vs One Zone-IA vs Glacier. Fix: learn each service by its one-line job AND by what it is NOT.
- **Qualifier words missed.** "Serverless", "without opening ports", "LEAST operational overhead", "immediately accessible", "choose how much traffic", "without manual configuration". These words decide the answer. Underline them before looking at the options.
- **Select-two questions.** In 7 of the 9 select-two questions you found one right answer but one of your picks was wrong; in 2 both picks were wrong. Judge every option on its own merit instead of picking a second one that merely looks plausible.
- **Exact facts that must be memorized:** 60-second EC2 minimum, Lambda 15-minute limit, S3 and Storage Gateway encrypted by default, Trusted Advisor idle-EC2 rule, the six CAF perspectives, shared controls.

Exam tip: When two options look similar, ask: "Which one does the exact verb in the question?" Schedule, deploy, scale, audit, monitor, recommend and share are all different verbs mapped to different services.

B. One-glance table of all 29 misses

Q	Topic	You picked (wrong)	Correct answer
1	Scheduled serverless job	Step Functions	EventBridge + Lambda
3	AMI and Regions	AMI is global	AMI must be in the same Region
4	Encrypted by default	Redshift	S3 + Storage Gateway
6	Migration expertise	APN Technology Partner	APN Consulting Partner
7	Right capacity for traffic	Application Load Balancer	EC2 Auto Scaling
14	Manage a fleet of nodes	AWS Health Dashboard	Systems Manager
16	CAF: strategy management	Operations	Business perspective
18	EC2 billing, 30-second run	30 seconds	60 seconds
22	Block-level storage	EFS	EBS + Instance Store
23	Always-free services	S3	IAM + Auto Scaling
25	RDS Read Replica benefit	Availability	Scalability

Q	Topic	You picked (wrong)	Correct answer
26	Prohibited uses policy	Fair Use Policy (made up)	Acceptable Use Policy
27	Unattached EBS volumes	CloudWatch	Trusted Advisor
30	Elastic Beanstalk	PaaS that models resources	PaaS to deploy and scale web apps
31	Account activity audit	Trusted Advisor	CloudTrail
34	Least-ops database	S3	DynamoDB
37	Share Reserved Instances	Systems Manager	AWS Organizations
38	Compute Optimizer scope	EC2 + EFS	EC2/ASG + EBS/Lambda
42	Rare but instant, recreatable	Glacier Flexible Retrieval	S3 One Zone-IA
45	USB-plug MFA	Hardware MFA device	U2F security key
47	Idle EC2, no setup	CUR + CloudWatch	Trusted Advisor + Cost Explorer
48	Cloud advantages	Opex for capex (backwards)	Speed/agility + stop guessing capacity
53	Shared Responsibility	Config mgmt is customer's	AWS 'of' cloud + abstracted services
57	Personalized status view	Service Health Dashboard	Your Account Health Dashboard
58	SSO for many accounts	Cognito	IAM Identity Center
61	Automate code deployment	CodePipeline	CodeDeploy
62	SSH, no open ports	EC2 Instance Connect	Systems Manager Session Manager
63	Route 53 traffic split	Latency-based	Weighted routing
64	CAF Platform stakeholders	CIO + CPO	Engineer + CTO

C. Question-by-question review

Compute, serverless and automation

Q1 | Scheduled serverless job	
QUESTION	Run a log backup every Monday at 2 AM (about 5 minutes). Recommend services for a serverless solution (select two).
YOU PICKED	**AWS Step Functions (plus Lambda, which was right)**
CORRECT	**Amazon EventBridge (Scheduler) + AWS Lambda**
WHY	A scheduled job needs two things: a trigger and something to run the code. EventBridge Scheduler is the serverless cron-style trigger; Lambda runs the code, and its 15-minute maximum covers a 5-minute job. Step Functions coordinates multi-step workflows but does not schedule by itself. Systems Manager is not the scheduler here, and EC2 is not serverless.
REMEMBER	**Schedule = EventBridge. Run code with no servers = Lambda. Lambda max runtime = 15 minutes.**

Q3 | AMI and Regions	
QUESTION	What Region-specific constraint must an AMI meet to launch an EC2 instance in a given Region?
YOU PICKED	**"An AMI is a global entity, so the Region is not applicable"**
CORRECT	**The AMI must be in the same Region as the instance. Its Region has no effect on performance.**
WHY	AMIs are Regional resources. If the AMI lives in another Region, copy it to the Region where you want to launch. The two "performance" options are made-up distractors.
REMEMBER	**AMI = regional. Need it elsewhere? Copy it. (Global services to remember: IAM, Route 53, CloudFront.)**

Q7 | Right amount of capacity	
QUESTION	Which entity makes sure an EC2 application always has the right capacity for current traffic?
YOU PICKED	**Application Load Balancer**
CORRECT	**Amazon EC2 Auto Scaling**
WHY	A load balancer only spreads traffic across targets that already exist. Auto Scaling adds and removes instances (within the min/max of an Auto Scaling group). Multi-AZ gives availability, not scaling.
REMEMBER	**Spread traffic = ELB. Match capacity to demand = Auto Scaling. Survive an AZ failure = Multi-AZ.**

Q14 | Manage a fleet of nodes

QUESTION	Centrally view, manage and operate nodes to quickly identify issues affecting applications.
YOU PICKED	**AWS Health Dashboard - Your Account Health**
CORRECT	**AWS Systems Manager**
WHY	Systems Manager (via the SSM Agent) lets you see and operate EC2 instances, on-premises servers and other clouds from one place: patching, automation, commands, sessions. The Health Dashboard only tells you about AWS-side events. Inspector scans for vulnerabilities. Trusted Advisor recommends best practices.
REMEMBER	**"Operate and manage a fleet of servers" = Systems Manager.**

Q30 | Elastic Beanstalk

QUESTION	Which statement MOST accurately describes AWS Elastic Beanstalk?
YOU PICKED	**"A PaaS that allows you to model and provision resources needed for an application"**
CORRECT	**A Platform as a Service (PaaS) that allows you to deploy and scale web applications and services**
WHY	You upload code; Beanstalk handles capacity provisioning, load balancing, auto scaling and health monitoring, and you manage only the application and data. "Model and provision resources" is the definition of CloudFormation (Infrastructure as Code). Beanstalk is not IaaS because you do not manage the runtime, middleware or OS.
REMEMBER	**Beanstalk = PaaS (deploy apps). CloudFormation = IaC (templates). EC2 = IaaS. Lambda = serverless.**

Q61 | Automate code deployment

QUESTION	Which service automates code deployment to EC2 instances and on-premises instances?
YOU PICKED	**AWS CodePipeline**
CORRECT	**AWS CodeDeploy**
WHY	CodeDeploy performs the actual deployment to EC2, on-premises servers and more, avoiding downtime. CodePipeline orchestrates the whole release process but calls CodeDeploy to do the deploying. CodeCommit hosts Git repositories. CloudFormation provisions infrastructure from templates.
REMEMBER	**Store code = CodeCommit. Build/test = CodeBuild. Deploy = CodeDeploy. Orchestrate all = CodePipeline.**

Q62 | SSH without open ports	
QUESTION	Provide secure shell access to EC2 instances without opening new ports or using public IP addresses.
YOU PICKED	**Amazon EC2 Instance Connect**
CORRECT	**AWS Systems Manager Session Manager**
WHY	Session Manager gives a browser or CLI shell with no inbound ports, no bastion hosts and no SSH keys to manage, and sessions are auditable. EC2 Instance Connect still needs port 22 open, so it fails the requirement. Inspector and Route 53 do not provide shell access.
REMEMBER	**"No open ports / no public IP / no SSH keys" = Session Manager.**

Storage and databases

Q4 | Encryption on by default	
QUESTION	Which services have data encryption automatically enabled? (select two)
YOU PICKED	**Amazon Redshift (plus S3, which was right)**
CORRECT	**Amazon S3 + AWS Storage Gateway**
WHY	All S3 buckets encrypt new objects by default (SSE-S3). Storage Gateway encrypts data in transit between the gateway and AWS using SSL. EBS, EFS and Redshift are NOT encrypted by default: encryption is an option you turn on (for EBS you can enforce it account-wide).
REMEMBER	**On automatically: S3 and Storage Gateway. You must enable it: EBS, EFS, Redshift.**

Q22 | Block-level storage	
QUESTION	Which services offer block-level storage? (select two)
YOU PICKED	**Amazon EFS (plus EBS, which was right)**
CORRECT	**Amazon EBS + Instance Store**
WHY	Both behave like disks attached to an instance. Instance Store is physically attached, very fast, but temporary: data is lost if the instance stops, fails or terminates, and it cannot be shared between instances. EFS is a shared file system (NFS), S3 is object storage, ECS is a container service.
REMEMBER	**Block = EBS and Instance Store. File = EFS (and FSx). Object = S3.**

Q25 | RDS Read Replica	
QUESTION	What is the primary benefit of deploying RDS with a Read Replica?
YOU PICKED	**Read Replica enhances database availability**
CORRECT	**Read Replica improves database scalability**
WHY	Read Replicas are read-only copies used to spread read traffic, which is horizontal scaling and better read performance. Availability and failover come from Multi-AZ. Replicas add cost, so "reduces cost" is wrong too.
REMEMBER	**Read Replica = scale reads. Multi-AZ = availability and failover.**

Q34 | Least-ops database	
QUESTION	Store recommendation-engine data with the LEAST operational overhead at any scale.
YOU PICKED	**Amazon S3**
CORRECT	**Amazon DynamoDB**
WHY	DynamoDB is a serverless NoSQL database: no provisioning, patching or cluster scaling. S3 is object storage, never a database. RDS needs more management at scale, and Neptune is for graph data.
REMEMBER	**"Database + least operational overhead + any scale" = DynamoDB. S3 is never the database answer.**

Q42 | Rare but instant, recreatable	
QUESTION	Thumbnails are rarely used but must be immediately accessible, and can be regenerated if lost. Cheapest S3 option?
YOU PICKED	**S3 Glacier Flexible Retrieval**
CORRECT	**S3 One Zone-Infrequent Access**
WHY	Glacier Flexible Retrieval takes minutes to hours to retrieve, so it breaks the "immediately accessible" need. One Zone-IA gives millisecond access at about 20% less than Standard-IA. Its single-AZ storage (lower availability) does not matter because the data can be recreated.
REMEMBER	**Rare + instant + recreatable = One Zone-IA. Rare + instant + must be resilient = Standard-IA.**

Security, identity and governance

Q45 | USB-plug MFA	
QUESTION	Which MFA mechanism supports a device you plug into a USB port?
YOU PICKED	**Hardware MFA device**
CORRECT	**U2F security key**
WHY	A U2F (FIDO) security key plugs into USB and you simply tap it. A hardware MFA device is a key-fob style token that shows a six-digit code you type in. A virtual MFA device is an authenticator app on a phone.
REMEMBER	**Plug in and tap = U2F. Code on a fob = hardware. Code on a phone app = virtual.**

Q58 | SSO across accounts	
QUESTION	Simplify access management to multiple AWS accounts and provide single sign-on.
YOU PICKED	**Amazon Cognito**
CORRECT	**AWS IAM Identity Center**
WHY	IAM Identity Center (successor to AWS SSO) gives your workforce one portal into many AWS accounts and SAML apps. Cognito is for sign-up and sign-in of your application's customers. IAM manages users, roles and policies but is not the SSO portal.
REMEMBER	**Employees into many AWS accounts = Identity Center. App end-customers = Cognito. Link to on-prem Active Directory = Directory Service.**

Q53 | Shared Responsibility Model	
QUESTION	Which statements are correct about the Shared Responsibility Model? (select two)
YOU PICKED	**"Configuration management is the responsibility of the customer" (plus 'AWS is responsible for security OF the cloud', which was right)**
CORRECT	**AWS is responsible for security 'of' the cloud + For abstracted services like S3, AWS operates the infrastructure layer, the OS and platforms**
WHY	Configuration management is a SHARED control: AWS configures its own infrastructure devices, you configure your guest OS, databases and applications. For EC2 you own the guest OS and patches. For abstracted services (S3, DynamoDB) AWS runs the infrastructure, OS and platform, and you just use the endpoints and protect your data and access.
REMEMBER	**Shared controls: patch management, configuration management, awareness and training.**

Q31 | Account activity for audit	
QUESTION	Ensure AWS account activity meets governance, compliance and auditing norms.
YOU PICKED	**AWS Trusted Advisor**
CORRECT	**AWS CloudTrail**
WHY	CloudTrail records who did what across the console, SDKs, CLI and other services, which is the audit trail compliance teams need. Config tracks resource configuration changes, CloudWatch monitors performance, and Trusted Advisor gives best-practice recommendations.
REMEMBER	**CloudWatch = performance and alarms. CloudTrail = account activity (who did what). Config = resource change history.**

Q57 | Personalized health view	
QUESTION	Which service gives a personalized view of the status of the AWS services in YOUR architecture?
YOU PICKED	**AWS Health - Service Health Dashboard**
CORRECT	**AWS Health - Your Account Health Dashboard**
WHY	Your Account Health Dashboard (formerly Personal Health Dashboard) shows events affecting your own resources and gives remediation guidance. The Service Health Dashboard only shows the general status of AWS services for everyone.
REMEMBER	**Personalized = Your Account Health. General public status = Service Health.**

Q26 | Prohibited uses policy	
QUESTION	Which policy describes prohibited uses of the web services offered by AWS?
YOU PICKED	**AWS Fair Use Policy**
CORRECT	**AWS Acceptable Use Policy**
WHY	The Acceptable Use Policy (AUP) lists what you may not do on AWS. "Fair Use Policy" and "Applicable Use Policy" are invented distractors, and Trusted Advisor is a recommendations tool.
REMEMBER	**AUP = Acceptable Use Policy = what is NOT allowed.**

Billing, cost and support

Q18 | EC2 minimum charge	
QUESTION	A Linux on-demand EC2 instance with per-second billing is terminated after 30 seconds. How long is it charged?
YOU PICKED	**30 seconds**
CORRECT	**60 seconds**
WHY	Per-second billing for Linux on-demand instances comes with a one-minute (60-second) minimum charge.
REMEMBER	**Per-second billing, 60-second minimum.**

Q23 | Always-free services	
QUESTION	Which services are always free to use? (select two)
YOU PICKED	**Amazon S3 (plus IAM, which was right)**
CORRECT	**AWS IAM + AWS Auto Scaling**
WHY	IAM and Auto Scaling carry no additional charge (you only pay for the resources Auto Scaling launches, and for CloudWatch if used). EC2, S3 and DynamoDB are paid services, billed on usage or storage.
REMEMBER	**Free features: IAM, Auto Scaling, Organizations (CloudFormation and Beanstalk are free too, but you pay for the resources they create).**

Q27 | Unattached EBS volumes	
QUESTION	Which service analyzes your infrastructure to identify unattached or underutilized EBS volumes?
YOU PICKED	**Amazon CloudWatch**
CORRECT	**AWS Trusted Advisor**
WHY	Trusted Advisor has built-in checks across Cost Optimization, Performance, Security, Fault Tolerance and Service Limits, including underused or unattached EBS volumes. CloudWatch only gives metrics and alarms you must configure yourself. Config tracks configuration, and Inspector assesses security.
REMEMBER	**"Find waste or best-practice gaps" = Trusted Advisor.**

Q37 | Share Reserved Instances	
QUESTION	Business units with separate AWS accounts under-use or over-use their Reserved Instances. Most cost-optimal fix?
YOU PICKED	**Use AWS Systems Manager to manage accounts and share the RIs**
CORRECT	**Use AWS Organizations to manage all accounts and share the Reserved Instances**
WHY	Organizations (consolidated billing) pools usage so unused Reserved Instance benefit in one account can apply to others in the same organization, and it is free. Cost Explorer only analyzes costs, Trusted Advisor only recommends, Systems Manager operates servers.
REMEMBER	**Share RI or Savings Plans discounts across accounts = Organizations.**

Q38 | Compute Optimizer scope	
QUESTION	Which resources does AWS Compute Optimizer give recommendations for? (select two)
YOU PICKED	**EC2 instances + EFS (plus EC2 + Auto Scaling groups, which was right)**
CORRECT	**EC2 instances + EC2 Auto Scaling groups, and EBS volumes + Lambda functions**
WHY	Compute Optimizer uses machine learning on historical utilization to right-size EC2 instances, Auto Scaling groups, EBS volumes and Lambda functions. It does not cover S3 or EFS.
REMEMBER	**Compute Optimizer = EC2, Auto Scaling groups, EBS, Lambda. Never S3 or EFS.**

Q47 | Idle EC2 with no setup	
QUESTION	Identify under-utilized EC2 instances off-the-shelf, with no manual configuration (select two).
YOU PICKED	**AWS Cost & Usage Report + Amazon CloudWatch**
CORRECT	**AWS Trusted Advisor + AWS Cost Explorer**
WHY	Trusted Advisor flags EC2 instances that ran in the last 14 days with daily CPU of 10% or less and network I/O of 5 MB or less on 4 or more days. Cost Explorer has built-in rightsizing recommendations. CUR is raw detailed billing data, while CloudWatch and Budgets would need alarms or targets set up first.
REMEMBER	**"Without manual configuration" = built-in recommendations: Trusted Advisor and Cost Explorer.**

Q48 | Advantages of the cloud	
QUESTION	Which are advantages of using the AWS Cloud? (select two)
YOU PICKED	**"Trade operational expense for capital expense" (plus 'Stop guessing about capacity', which was right)**
CORRECT	**Increase speed and agility + Stop guessing about capacity**
WHY	You trade capital expense for variable (operational) expense, so that option was backwards. Scaling is not limited, and AWS handles security OF the cloud, not security IN it.
REMEMBER	**Cloud trades CapEx for OpEx. Learn all six advantages (listed in Domain 1).**

Q6 | Who helps you migrate	
QUESTION	A multinational wants expert professional advice on migrating to and managing applications on AWS.
YOU PICKED	**APN Technology Partner**
CORRECT	**APN Consulting Partner**
WHY	Consulting Partners are professional-services firms that design, build, migrate and manage workloads on AWS. Technology Partners supply software, hardware or connectivity products. The Concierge Support Team handles billing and account questions for Enterprise Support, and Trusted Advisor is just a tool.
REMEMBER	**Need people to do or advise migration = Consulting Partner. Need a product = Technology Partner.**

Cloud concepts and networking

Q16 | CAF: strategy management	
QUESTION	Which AWS CAF perspective includes strategy management?
YOU PICKED	**Operations perspective**
CORRECT	**Business perspective**
WHY	Strategy management sits in the Business perspective, which is about making sure cloud investment serves business outcomes. Operations is about running workloads: observability, incident and change management.
REMEMBER	**Strategy, portfolio, innovation, product management = Business perspective.**

Q64 | CAF Platform stakeholders	
QUESTION	Which are common stakeholder roles for the AWS CAF Platform perspective? (select two)
YOU PICKED	**Chief Information Officer (CIO) + Chief Product Officer (CPO)**
CORRECT	**Engineer + Chief Technology Officer (CTO)**
WHY	The Platform perspective is about the technical, enterprise-grade cloud environment, so its stakeholders are the CTO, technology leaders, architects and engineers.
REMEMBER	**Platform perspective = technical people: CTO, architects, engineers.**

Q63 | Route 53 routing policy	
QUESTION	Route traffic to multiple resources and choose how much traffic goes to each.
YOU PICKED	**Latency-based routing**
CORRECT	**Weighted routing**
WHY	Weighted routing assigns a relative weight to each record, so you control the share of traffic per resource (handy for testing a new version on 10% of traffic). Latency-based picks the Region with the best latency, failover is active-passive, and simple routing points to one resource.
REMEMBER	**"Choose how much traffic to each" = Weighted. "Closest / fastest Region" = Latency.**

D. Revision tables for the topics you missed

1. AWS Cloud Adoption Framework (CAF): the six perspectives

Perspective	Focus	Keywords and example capabilities
Business	Cloud investment should drive business outcomes	Strategy management, portfolio management, innovation management, product management, data monetization
People	Culture, skills and organizational change	Culture evolution, leadership, cloud fluency, workforce transformation, change acceleration
Governance	Align cloud with business goals, manage risk and cost	Program and project management, risk management, cloud financial management, benefits management
Platform	Build the technical cloud environment	Platform architecture, platform engineering, data architecture, provisioning, CI/CD. Stakeholders: CTO, architects, engineers
Security	Visibility, auditability, control and agility for security	Identity and access management, threat detection, vulnerability management, data protection, incident response
Operations	Run workloads reliably day to day	Observability, incident and problem management, change and release management, configuration and patch management

Exam tip: Memory hook: **B**usiness = strategy. **P**eople = skills and culture. **G**overnance = risk and money. **P**latform = tech build. **S**ecurity = protect. **O**perations = keep it running.

2. RDS: Multi-AZ vs Read Replica vs Multi-Region

Option	Main benefit	Details and trigger words
Multi-AZ	High availability and automatic failover	Synchronous standby in another AZ, same Region. "Availability", "failover", "durability"
Read Replica	Read scalability and performance	Asynchronous, read-only copies. "Scale reads", "offload reporting", horizontal scaling. Adds cost
Cross-Region replica / Multi-Region	Disaster recovery and lower latency for distant users	Survives a whole-Region failure. "Regional failure", "globally redundant"

Note: your two practice sources phrase this slightly differently. If the question asks for the *primary benefit of a Read Replica*, answer scalability. If it asks which RDS feature creates globally redundant databases, the cross-Region replica is the answer. Read the exact wording.

3. Route 53 routing policies

Policy	Use it when
Simple	One resource for a domain (a single web server).
Weighted	You want to control the proportion of traffic per resource (load balancing, canary testing a new version).
Latency-based	Resources in several Regions and you want users sent to the lowest-latency Region.
Failover	Active-passive setup: send traffic to the standby only if the primary is unhealthy.

Policy	Use it when
Geolocation	Route by where the user is located (country or continent), for localization or legal rules.

4. Storage types: block vs file vs object

Type	Services	Key facts
Block	EBS, Instance Store	Disk-like, attached to an EC2 instance. EBS persists independently. Instance Store is temporary and lost on stop, failure or termination.
File	EFS, FSx	Shared file system accessible by many instances at once (EFS = NFS).
Object	S3 (and Glacier classes)	Objects accessed by key over an API/URL. Not a database, not a disk.

5. S3 storage classes: pick by access pattern

Class	Use when
Standard	Frequently accessed data, stable access (websites, active content).
Intelligent-Tiering	Unknown or changing access patterns; moves data between tiers for you.
Standard-IA	Infrequent access but rapid retrieval; stored across multiple AZs; retrieval fee.
One Zone-IA	Infrequent, rapid access, data can be recreated; single AZ, about 20% cheaper than Standard-IA.
Glacier Instant Retrieval	Archive data rarely accessed but needed in milliseconds.
Glacier Flexible Retrieval	Archive data; retrieval takes minutes to hours.
Glacier Deep Archive	Lowest cost; long-term compliance archives; retrieval takes hours.

6. Encryption: on by default or not?

Service	Default behavior
Amazon S3	Encrypted by default (SSE-S3) for all new objects.
AWS Storage Gateway	Data in transit between the gateway and AWS is encrypted with SSL.
Amazon EBS	Not by default. Can be enabled, or enforced for all new volumes in the account.
Amazon EFS	Optional (in transit and at rest).
Amazon Redshift	Optional setting.

7. MFA device types

Type	How it works
Virtual MFA	Authenticator app on a phone or other device generating a six-digit time-based code.
Hardware MFA	Physical token (fob) that displays a six-digit code you type in.

Type	How it works
U2F / FIDO security key	Plugs into a USB port; you tap it instead of typing a code.
SMS-based MFA	Code sent by text message to the registered phone (listed in older exam questions).

8. Identity services: who is it for?

Service	Use for
IAM	Users, groups, roles and policies inside an AWS account. Free.
IAM Identity Center	Workforce single sign-on into multiple AWS accounts and SAML apps (successor to AWS SSO).
Amazon Cognito	Sign-up and sign-in for the customers of your web or mobile app.
AWS Directory Service	Managed Microsoft Active Directory, or connecting to on-premises AD.

9. Developer tools (CI/CD family)

Service	Job
CodeCommit	Git-based source control repositories.
CodeBuild	Compiles code and runs tests.
CodeDeploy	Automates deployment to EC2, on-premises servers and more.
CodePipeline	Orchestrates the whole release workflow and calls the other tools (it does not deploy by itself).
CloudFormation	Infrastructure as Code: provisions resources from templates.
Elastic Beanstalk	PaaS: upload app code, AWS handles provisioning, load balancing, scaling and health.

10. Monitoring, auditing, recommending: the confusion cluster

Service	Think of it as	Question clue
CloudWatch	Performance monitoring, metrics, alarms	"Monitor performance", "alert when CPU is high"
CloudTrail	Who did what (API/account activity)	"Audit", "governance", "user actions"
Config	Resource configuration history and compliance	"Changes made to a resource over time"
Trusted Advisor	Built-in best-practice checks (5 categories)	"Idle or unattached", "recommendations", "no setup"
Systems Manager	Operate and manage a fleet of servers	"Patch", "centrally manage nodes", "no open ports shell"
Health Dashboard	AWS service events (Your Account = personalized)	"Is AWS having an issue affecting me?"

Service	Think of it as	Question clue
Inspector	Vulnerability scanning	"Assess EC2 for vulnerabilities"

11. Shared Responsibility: the three kinds of control

Kind	Examples
Inherited (you inherit from AWS)	Physical and environmental controls.
Shared (both sides, different scope)	Patch management (AWS patches infrastructure, you patch guest OS and apps), configuration management, awareness and training.
Customer-specific	Controls you own entirely, such as service and communications protection, and your data and access settings.

For EC2 (IaaS) you manage the guest OS, patches, applications and security group. For abstracted services like S3 and DynamoDB, AWS runs the infrastructure, OS and platform, and you manage your data and access.

12. Who can help me with migration and account issues?

Entity	What it is
APN Consulting Partner	Professional-services firm that designs, builds, migrates and manages workloads.
APN Technology Partner	Provides software, hardware or connectivity that runs on or integrates with AWS.
AWS Professional Services	AWS's own team that works with you on adoption and migration.
Concierge Support Team	Billing and account experts, available only with Enterprise Support.
AWS Acceptable Use Policy	The rules on what you must not do on AWS.

13. Billing and cost facts that showed up

- EC2 on-demand: per-second billing with a **60-second minimum**.
- Always free (no extra charge): IAM, Auto Scaling, Organizations. You still pay for the resources they manage.
- Share Reserved Instance or Savings Plan benefit across accounts: **AWS Organizations** (consolidated billing).
- Trusted Advisor idle-EC2 rule: ran in the last 14 days, daily CPU 10% or less and network I/O 5 MB or less on 4 or more days.
- Cost Explorer includes rightsizing recommendations; Compute Optimizer covers EC2, Auto Scaling groups, EBS and Lambda.

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*Redo the 29 questions above after a day or two without looking at the answers. If you can explain **why each wrong option is wrong**, you have fixed the pattern, not just memorized the answer.*